Name _____ Period _____ Date _____

2-2 Lesson Quiz: Standard Form of a Quadratic Function

short form

2. The amount of profit Bill makes per toy when he increases or decreases the price of his handmade toys can be modeled by the function $f(x) = -x^2 - 2x + 3$. What price change gives him the highest profit? What is his highest profit per toy?

Price change: $ ☐

Maximum profit per toy: $ ☐

Show your work:

_____

_____

_____

3. A pebble is tossed into the air from the top of a cliff. The height, in feet, of the pebble over time is modeled by the equation $y = -16x^2 + 32x + 80$. What is the maximum height, in feet, reached by the pebble?

Maximum height: ☐ ft

Show your work:

_____

_____

_____

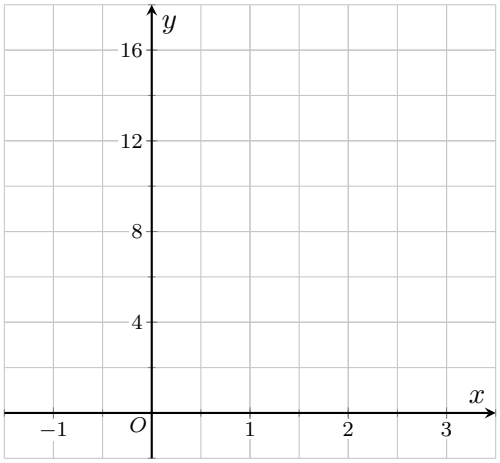
5. An equation that represents the path of a diver jumping off a diving board is $y = -7x^2 + 5x + 16$. Graph this function from the point where the diver jumped to the point where they reach the water.

Show your thinking:

Where the diver jumped: (_____ , _____)

Vertex: (_____ , _____)

Reaches the water: about (_____ , _____)



Scoring: Question 3 and Question 5 are 10 points each. Question 2 is a bonus of up to 5 points (2 for the price change, 3 for the profit). Total out of 20. (Question numbers are the Savvas quiz numbers; items 1 and 4 are on the full Savvas quiz only.)

2-2 Lesson Quiz, short form: answer key

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2. **Price change: \$−1. Maximum profit per toy: \$4.** $a = -1$, $b = -2$: $h = -\dfrac{b}{2a} = -\dfrac{(-2)}{2(-1)} = -1$, and $f(-1) = -1 + 2 + 3 = 4$. So lowering the price by \$1 gives the highest profit, \$4 per toy. Watch for $h = +1$ (the sign of b dropped; the Common Error the TE flags for Try It 2), which gives $f(1) = 0$. Savvas quiz item 2, DOK 2, skill F16 (Example 3). **Bonus 5:** 2 for -1 (accept "decrease the price by \$1"), 3 for \$4.

3. **96 ft.** $h = -\dfrac{32}{2(-16)} = 1$, and $y = -16(1)^2 + 32(1) + 80 = 96$. Savvas quiz item 3, DOK 2, skill F16 (Example 3; Try It 3 is the same task). **10 points:** 4 for $h = 1$ with the formula shown, 4 for substituting $x = 1$ correctly, 2 for 96 ft.

5. **A downward parabola segment from** $(0, 16)$ **to about** $(1.91, 0)$**, vertex about** $(0.36,\ 16.89)$**; closed endpoints, no arrows.** $h = -\dfrac{5}{2(-7)} = \dfrac{5}{14} \approx 0.36$; $k = -7\left(\frac{5}{14}\right)^2 + 5\left(\frac{5}{14}\right) + 16 = 16\frac{25}{28} \approx 16.89$. The symmetric point of $(0, 16)$ is $\left(\frac{5}{7}, 16\right) \approx (0.71, 16)$. A table gives $y(1) = 14$, $y(1.5) = 7.75$, $y(2) = -2$, so the diver reaches the water between $x = 1.5$ and $x = 2$, near $x \approx 1.91$. Savvas quiz item 5, DOK 2, skill F16 (Example 2 with Example 3's context). **10 points, all for the graph the TE asks for:** 2 for starting at $(0, 16)$, 3 for a vertex near $(0.36, 16.9)$, 3 for ending near $(1.9, 0)$ (accept 1.8–2.0), 2 for drawing only $0 \le x \le 1.91$ with closed ends. The three "show your thinking" blanks are the same points written down; they earn these points when the graph is missing or unclear.

Items left off this form (full Savvas quiz only; available for a retake or Form B): Savvas item 1, the minimum point of $f(x) = x^2 - 4x$ (answer D, $(2, -4)$; Example 1/2 vertex skill, DOK 2, F16); Savvas item 4, the parabola through $(-5, -10)$, $(-1, 6)$ and $(2, -3)$ (answer A, $y = -x^2 - 2x + 5$; Example 4, not taught in the fast packet, DOK 2, F50; students can still check each choice by substituting the three points).

IXL replacement: SmartScore 70 or above on L8C and S9G replaces this quiz with 20/20.